The matrix exponential

General definition of the matrix exponential (via power series expansion):

$$\mathrm{e}^A \equiv \exp(A) = \sum_{k=0}^{\infty} \frac{1}{k!} A^k, \quad A \in \mathbb{C}^{n\times n}$$

with $A^k = A \cdot A \cdots A$ (k times) and $n \in \mathbb{N}$.

Generalization of Euler's formula Special case: self-inverse matrix $A^2 = I$ (identity), and $x \in \mathbb{R}$ a real number. Then

$$\mathrm{e}^{iAx} = \cos(x)I + i\sin(x)A,$$

which generalizes Euler's formula $\mathrm{e}^{ix} = \cos(x) + i\sin(x)$.

Proof. Since $A^2 = I$, it is clear that $A^k = \begin{cases} I & k \text{ even} \\ A & k \text{ odd} \end{cases}$. Therefore, the power series expansion of the exponential can be partitioned as

$$\begin{aligned} \mathrm{e}^{iAx} &= I + iAx - \frac{1}{2!}Ix^2 - \frac{i}{3!}Ax^3 + \dots \\ &= I\left(1 - \frac{1}{2!}x^2 + \dots\right) + iA\left(x - \frac{1}{3!}x^3 + \dots\right). \end{aligned}$$

Note that the series inside the brackets correspond to the Taylor expansions of $\cos$ and $\sin$:

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

In summary,

$$\mathrm{e}^{iAx} = \cos(x)I + i\sin(x)A.$$

□

Unitary matrix exponentials If $A \in \mathbb{C}^{n\times n}$ ($n \in \mathbb{N}$) is Hermitian, then e^{iA} is unitary.

A is called Hermitian if $A^\dagger = A$, with $A^\dagger = (A^*)^T$ denoting the conjugate transpose. $U \in \mathbb{C}^{n\times n}$ is called unitary if $U^\dagger U = I$.

Proof. We first observe that

$$\left(\mathrm{e}^{iA}\right)^\dagger = \sum_{k=0}^{\infty} \frac{1}{k!} \left((iA)^k\right)^\dagger = \sum_{k=0}^{\infty} \frac{1}{k!} \left(-iA^\dagger\right)^k = \sum_{k=0}^{\infty} \frac{1}{k!} (-iA)^k = \mathrm{e}^{-iA}.$$

For any square matrices A and B which *commute* ($[A, B] = AB - BA = 0$), it holds that $\mathrm{e}^A \mathrm{e}^B = \mathrm{e}^{A+B}$. (This can be shown analogous to the scalar case, but does not hold in general if $[A, B] \neq 0$.) Thus

$$\left(\mathrm{e}^{iA}\right)^\dagger \mathrm{e}^{iA} = \mathrm{e}^{-iA}\,\mathrm{e}^{iA} = \mathrm{e}^{-iA+iA} = \mathrm{e}^0 = I.$$

Here we have used that $-iA$ commutes with iA (since these matrices only differ by a sign factor). □

Single-qubit rotation operators Pauli matrices X, Y, Z are Hermitian and self-inverse. Can be used to define the following unitary *rotation operators*, with parameter $\theta \in \mathbb{R}$:

$$R_x(\theta) \coloneqq \mathrm{e}^{-i\theta X/2} = \cos(\theta/2)I - i\sin(\theta/2)X = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

$$R_y(\theta) \coloneqq \mathrm{e}^{-i\theta Y/2} = \cos(\theta/2)I - i\sin(\theta/2)Y = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

$$R_z(\theta) \coloneqq \mathrm{e}^{-i\theta Z/2} = \cos(\theta/2)I - i\sin(\theta/2)Z = \begin{pmatrix} \mathrm{e}^{-i\theta/2} & 0 \\ 0 & \mathrm{e}^{i\theta/2} \end{pmatrix}$$

General case: rotation about an axis $\vec{v} \in \mathbb{R}^3$ (normalized such that with $\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2} = 1$):

Notation

$$\underbrace{\vec{v} \cdot \vec{\sigma}}_{\langle \vec{v}|\vec{\sigma}\rangle} \equiv v_1 X + v_2 Y + v_3 Z = \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix} \in \mathbb{C}^{2\times 2}$$

It holds that $\vec{v} \cdot \vec{\sigma}$ is Hermitian and $(\vec{v} \cdot \vec{\sigma})^2 = I$:

$$\begin{aligned}(\vec{v} \cdot \vec{\sigma})^2 &= \begin{pmatrix} v_3^2 + (v_1 - iv_2)(v_1 + iv_2) & v_3(v_1 - iv_2) + (v_1 - iv_2)(-v_3) \\ (v_1 + iv_2)v_3 - v_3(v_1 + iv_2) & v_3^2 + (v_1 - iv_2)(v_1 + iv_2) \end{pmatrix} \\ &= \begin{pmatrix} v_3^2 + v_1^2 + v_2^2 & 0 \\ 0 & v_3^2 + v_1^2 + v_2^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},\end{aligned}$$

where in the last step we have used that $\vec{v}$ is a unit vector, and therefore $\|\vec{v}\|^2 = v_1^2 + v_2^2 + v_3^2 = 1$.

General unitary *rotation operator about axis* $\vec{v}$ with angle $\theta \in \mathbb{R}$:

$$R_{\vec{v}}(\theta) \coloneqq \mathrm{e}^{-i\theta(\vec{v}\cdot\vec{\sigma})/2} = \cos(\theta/2)I - i\sin(\theta/2)(\vec{v} \cdot \vec{\sigma}).$$

Note: R_x, R_y, R_z are special cases, corresponding to $\vec{v} = (1, 0, 0)$, $\vec{v} = (0, 1, 0)$, $\vec{v} = (0, 0, 1)$.

Can derive that Bloch sphere representation of $R_{\vec{v}}(\theta)$ is a "conventional" rotation (in three dimensions) by angle θ about axis $\vec{v}$ (see Tutorial 11): with $\vec{r}$ the Bloch vector of $|\psi\rangle$,

$$|\psi'\rangle = R_{\vec{v}}(\theta)\,|\psi\rangle$$

